

HDOC - Day 24

Menu-driven Increasing Patterns

Algorithm

1. Start and display the menu containing the five increasing patterns and Exit.
2. Read the choice and select the required pattern using switch-case.
3. Read n inside the selected parameterless function.
4. Use nested loops: the outer loop controls rows and the inner loop prints the required symbols/numbers/letters.
5. Return to the menu and repeat until Exit is selected.

Test Case Table

Test	Operation	Input	Expected output
1	Increasing star triangle	n = 4	1, 2, 3, 4 stars by row
2	Row-number triangle	n = 4	1; 2 2; 3 3 3; 4 4 4 4
3	Natural-number triangle	n = 4	Rows print 1 through row no.
4	Consecutive-number triangle	n = 4	Numbers 1 through 10 triangularly
5	Increasing alphabet	n = 4	A through row letter

C Program

```
#include <stdio.h>
void stars(){int n,i,j;printf("Enter n: ");scanf("%d",&n);for(i=1;i<=n;i++){for(j=1;j<=i;j++)printf(" ");printf("\n");}}
void rows(){int n,i,j;printf("Enter n: ");scanf("%d",&n);for(i=1;i<=n;i++){for(j=1;j<=i;j++)printf("%d ",i);printf("\n");}}
void natural(){int n,i,j;printf("Enter n: ");scanf("%d",&n);for(i=1;i<=n;i++){for(j=1;j<=i;j++)printf("%d ",j);printf("\n");}}
void consecutive(){int n,i,j,k=1;printf("Enter n: ");scanf("%d",&n);for(i=1;i<=n;i++){for(j=1;j<=i;j++)printf("%d ",k++);printf("\n");}}
void alphabet(){int n,i,j;printf("Enter n (1-26): ");scanf("%d",&n);for(i=1;i<=n;i++){for(j=1;j<=i;j++)printf("%c ", 'A'+j-1);printf("\n");}}
int main(){int ch;do{printf("\n1. Increasing star triangle\n2. Row-number triangle\n3. Natural-number triangle\n4. Consecutive-number triangle\n5. Increasing alphabet\n6. Exit\n");printf("Enter choice: ");scanf("%d",&ch);switch(ch){case 1:stars();break;case 2:rows();break;case 3:natural();break;case 4:consecutive();break;case 5:alphabet();break;case 6:break;default:printf("Invalid choice\n");}}while(ch!=6);}
```

Execution

Test Case 1 - Increasing Right-angled Triangle

```
puppie@Anvikas-MacBook-Air Desktop % clang hdoc24.c -o hdoc24
puppie@Anvikas-MacBook-Air Desktop % ./hdoc24
Enter choice: 1
Enter n: 4
*
* *
* * *
* * * *
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 2 - Increasing Row-number Triangle

```
puppie@Anvikas-MacBook-Air Desktop % ./hdoc24
Enter choice: 2
Enter n: 4
1
2 2
3 3 3
4 4 4 4
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 3 - Increasing Natural-number Pattern

```
puppie@Anvikas-MacBook-Air Desktop % ./hdoc24
Enter choice: 3
Enter n: 4
1
1 2
1 2 3
1 2 3 4
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 4 - Consecutive-number Triangle

```
puppie@Anvikas-MacBook-Air Desktop % ./hdoc24
Enter choice: 4
Enter n: 4
1
2 3
4 5 6
7 8 9 10
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 5 - Increasing Alphabet Pattern

```
puppie@Anvikas-MacBook-Air Desktop % ./hdoc24
Enter choice: 5
Enter n (1-26): 4
A
A B
A B C
A B C D
puppie@Anvikas-MacBook-Air Desktop %
```

Result: The menu-driven C program using switch-case and nested loops was implemented and tested successfully for all five increasing patterns.